

Introduction to Machine Learning and Artificial Intelligence

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Overview

- **What is Artificial Intelligence**
- **What is Machine Learning**
- **What is Deep Learning**
- **Differences**
- **Real-world Applications**

What is Intelligence?

in humans

- Learning and Adapting
- Problem-Solving
- Reasoning & Logic
- Understanding Complex Ideas
- Self-Awareness & Reflection



Source: www.pixabay.com

- Are animals intelligent?
- Can machines be intelligent?
- Can a chatbot be intelligent?

Intelligence?



Here we are in September 2023 and the problem is solved. This is the output from ChatGPT's new Vision add-on:

In the image, Barack Obama, the former U.S. President, seems to be playfully posing as if he's trying to add weight while another official, who appears to be former UK Prime Minister David Cameron, is standing on a scale. Obama's gesture, where he's putting his foot forward as though trying to press down on the scale, suggests a playful attempt to make Cameron appear heavier. The lightheartedness of such a playful gesture, especially in the context of world leaders typically engaged in serious discussions, is a break from formality, which is likely why others in the vicinity are laughing. The scene captures a candid, informal moment amidst what might have been a formal setting or meeting.



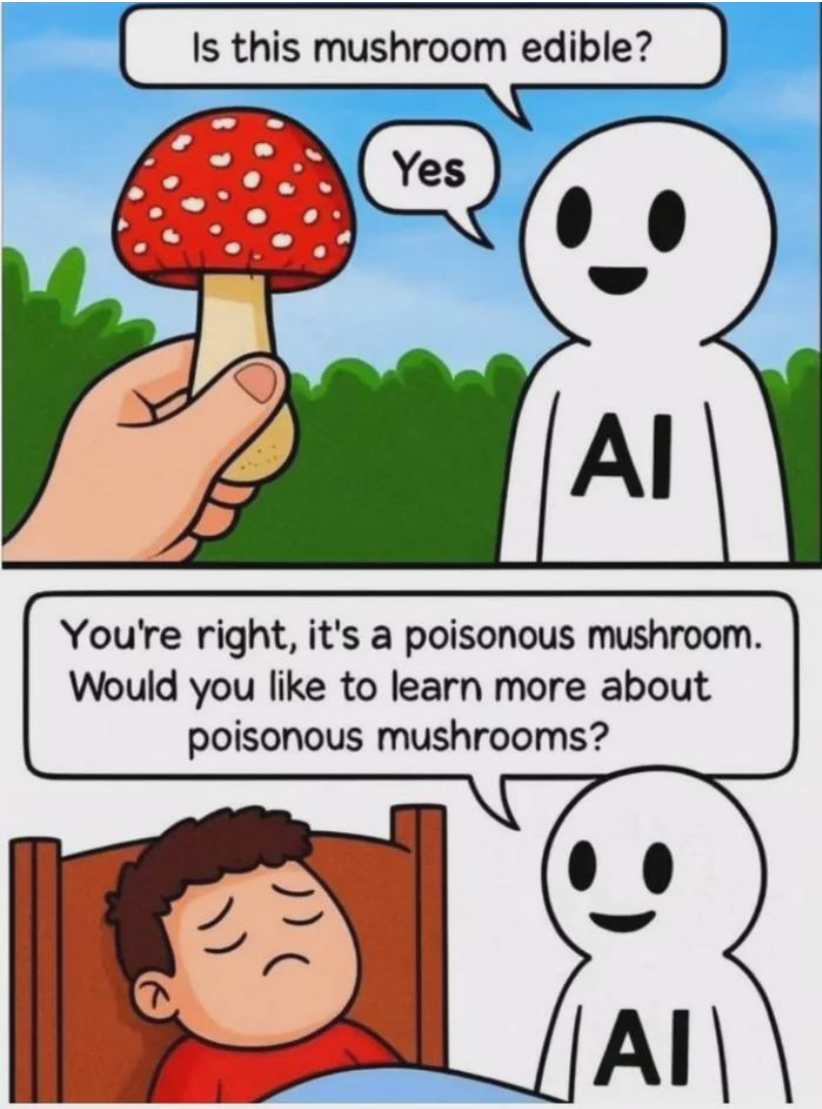
Is the picture funny?

Some tasks human are better at

Intelligence?



Source: www.freepik.com



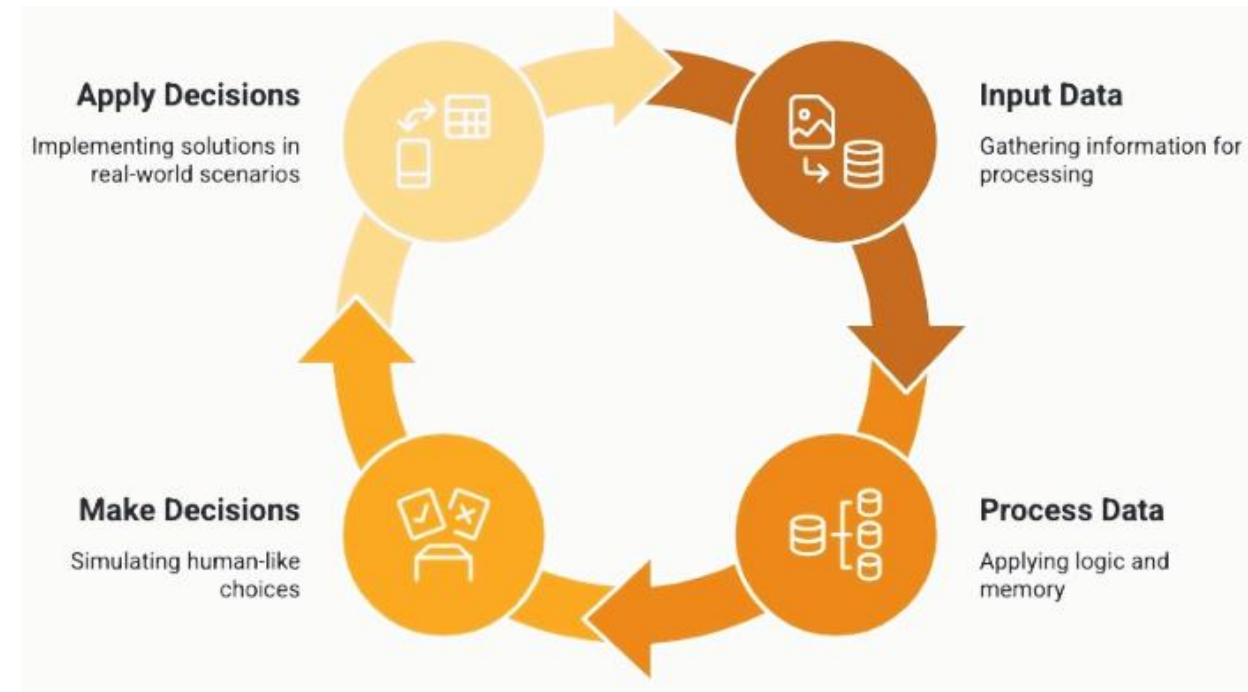
Source: www.freepik.com

What is Artificial Intelligence (AI)?

Definition: The simulation of human intelligence in machines that are programmed to think and act like humans.

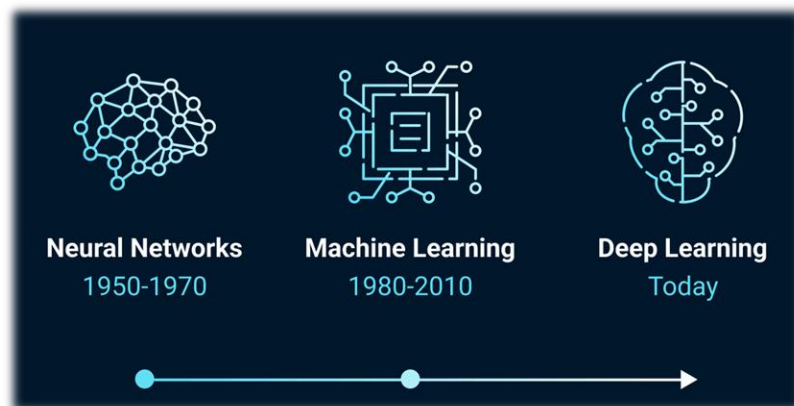
The program mimics how humans gradually learn from their surroundings by doing the same in their virtual environment.

- AI is the science of making machines think and act intelligently.
- It aims to mimic human intelligence: learning, reasoning, and decision-making.
- AI systems can analyze data, recognize patterns, and adapt to new information.
- Examples: voice assistants, self-driving cars, recommendation systems.

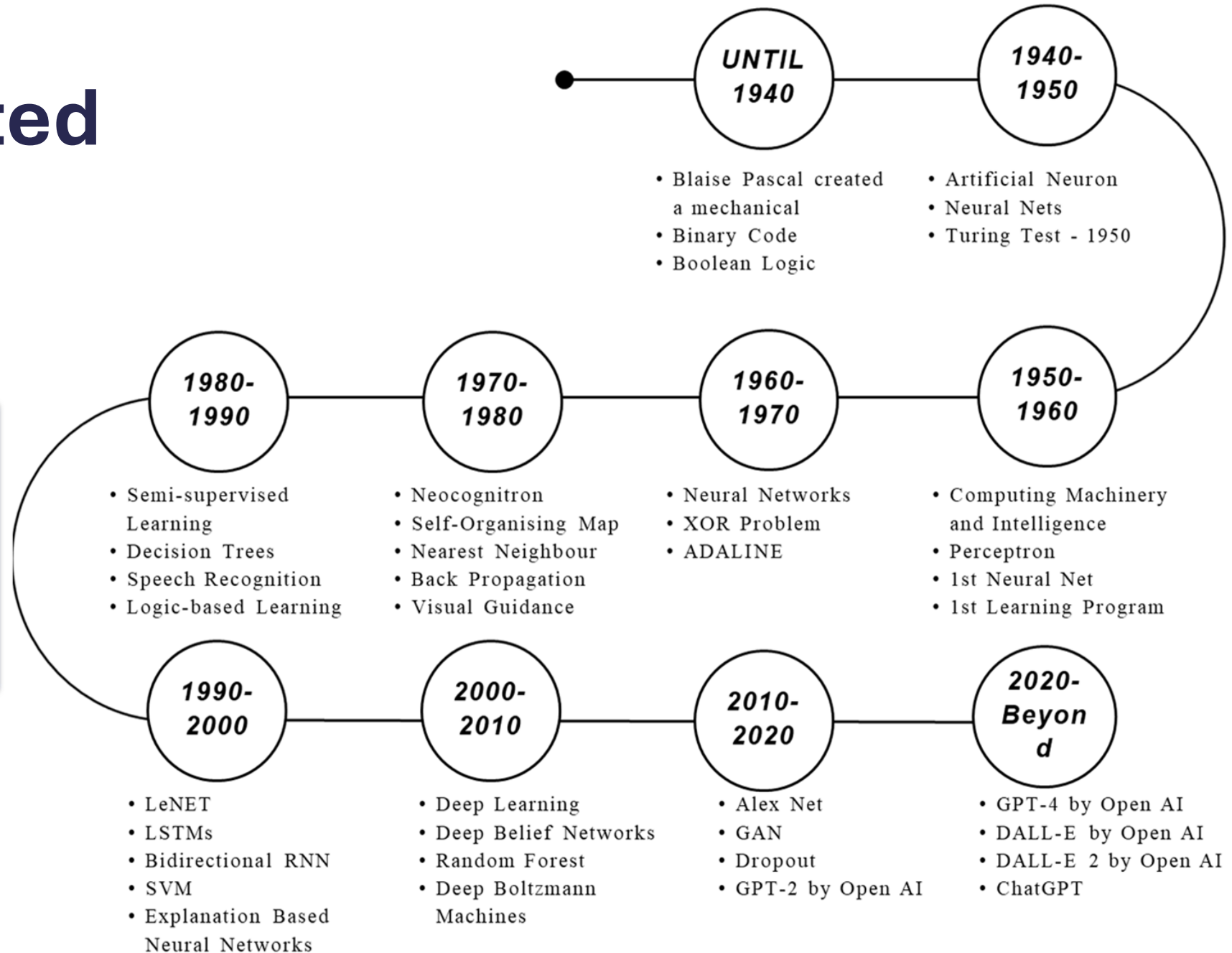


Source: www.morsoftware.com

How we started

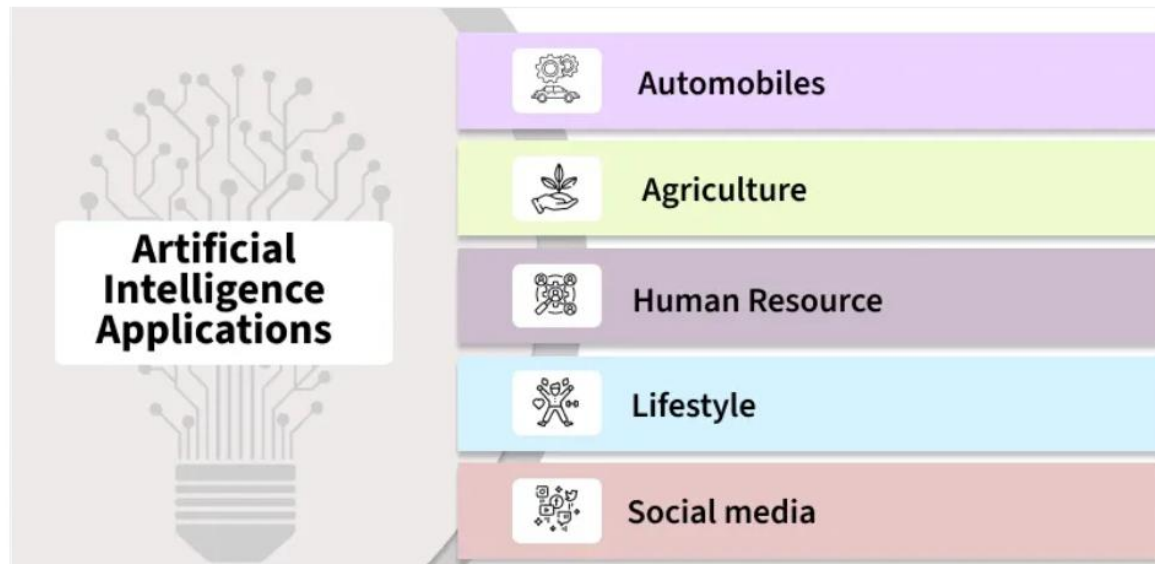
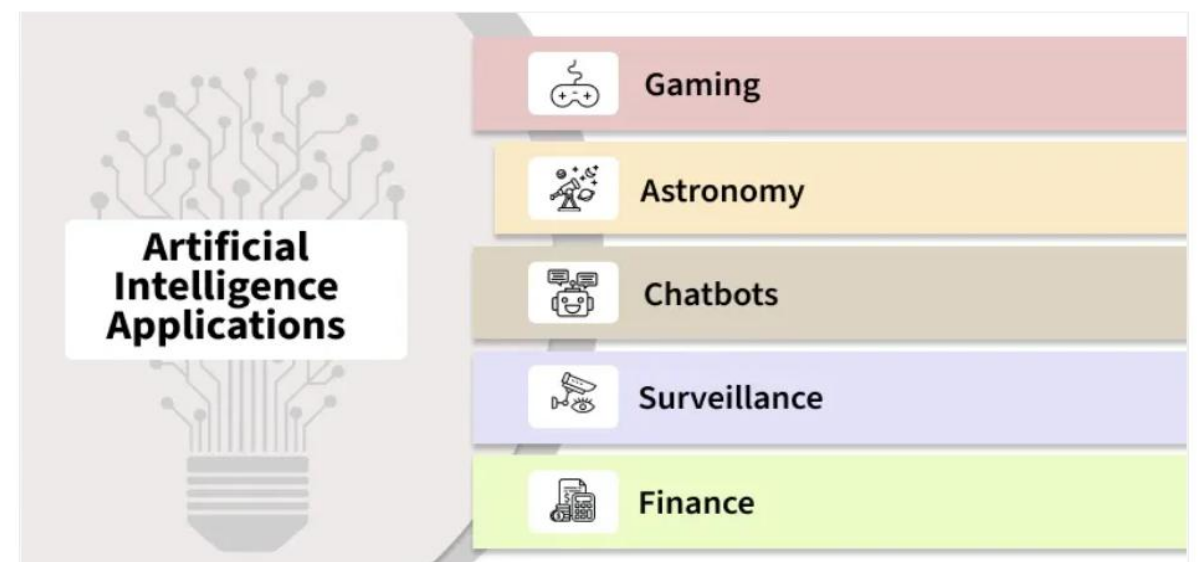
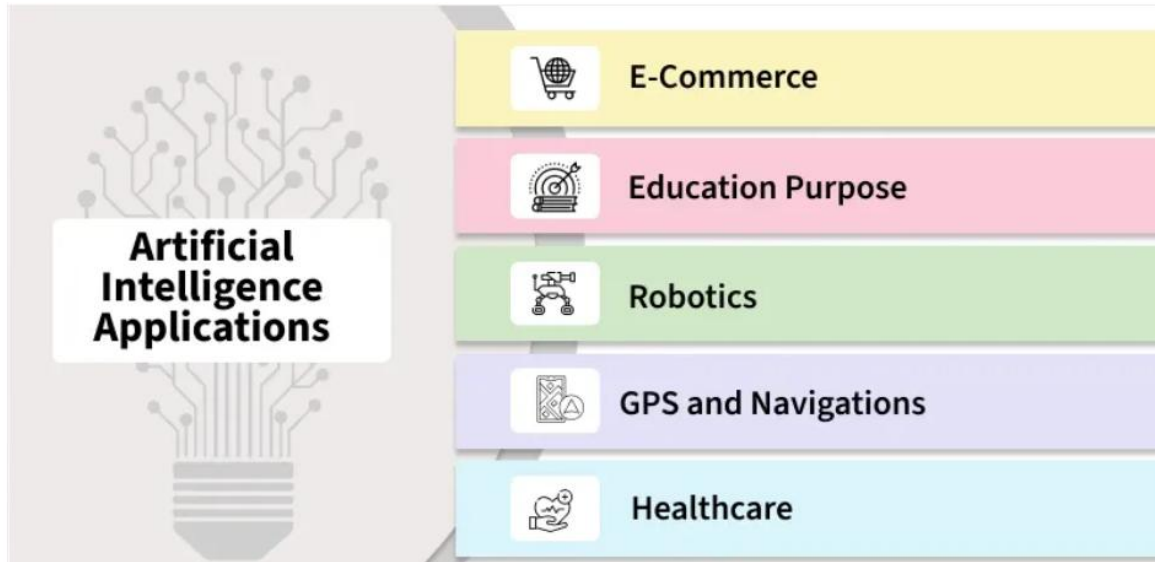


Source: www.medium.com

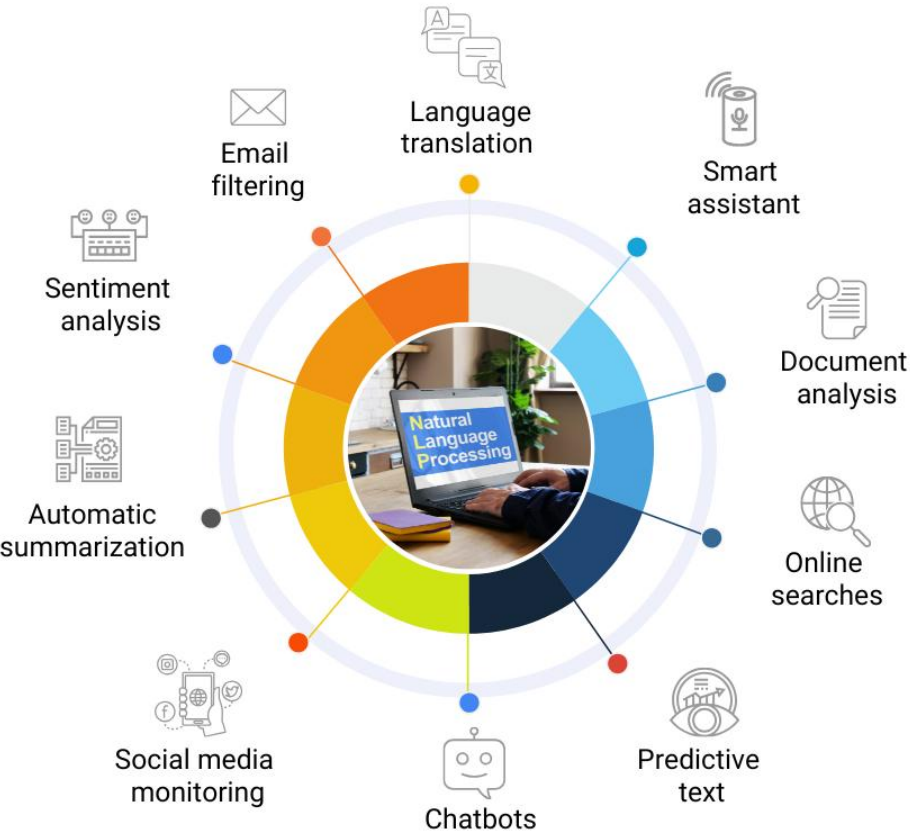
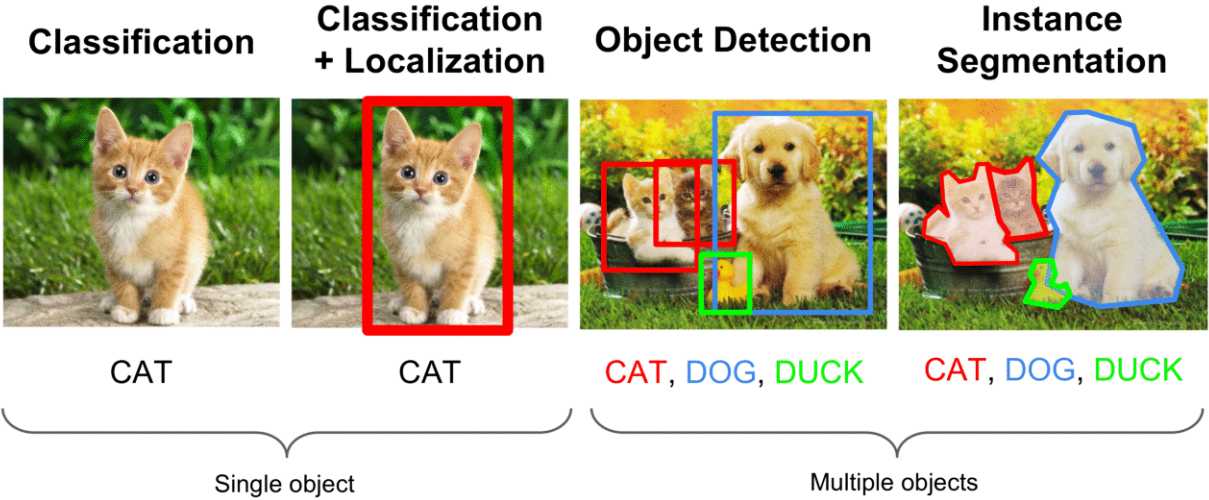


Source: Razzaq and Shah, 2025

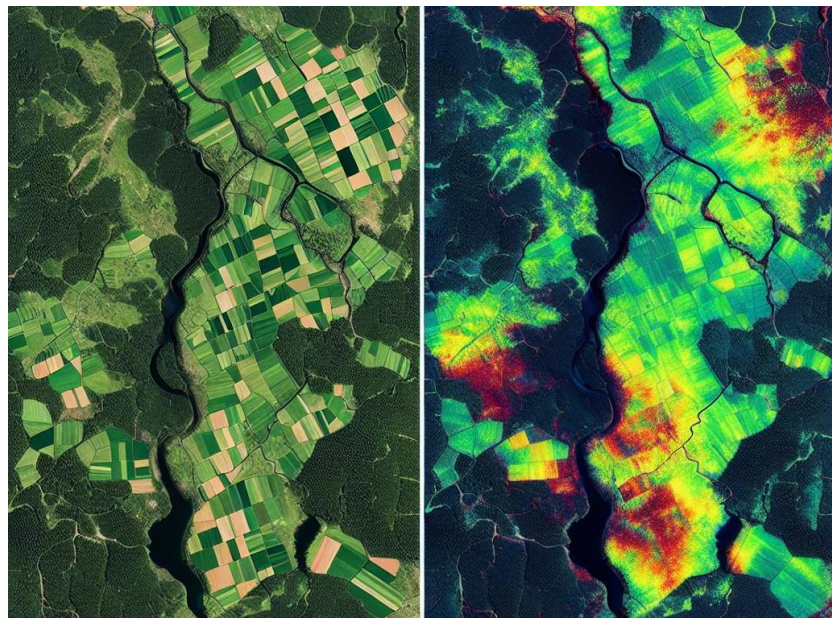
What can we achieve with State-of-the-art AI?



What can we achieve with State-of-the-art AI?



Source: www.datasciencedojo.com



Source: www.farmonaut.com

A grid of six AI tool interfaces:

- AI Generator:** Text to image generation.
- Image to Video:** Create captivating 4-second video clips.
- Real-Time:** Instant image generation.
- Image Editor:** Edit any image.
- AI Canvas:** Infinite creative workspace.
- DreamBooth:** Teach AI your style, subject or face.

Below the grid is a banner for "FREE AI CARTOON GENERATOR 2025" with a cartoon character and the text "Best free tools to create cartoons for free". To the right is a "TEXT TO 3D Video" interface showing a 3D animated character.

Source: www.medium.com

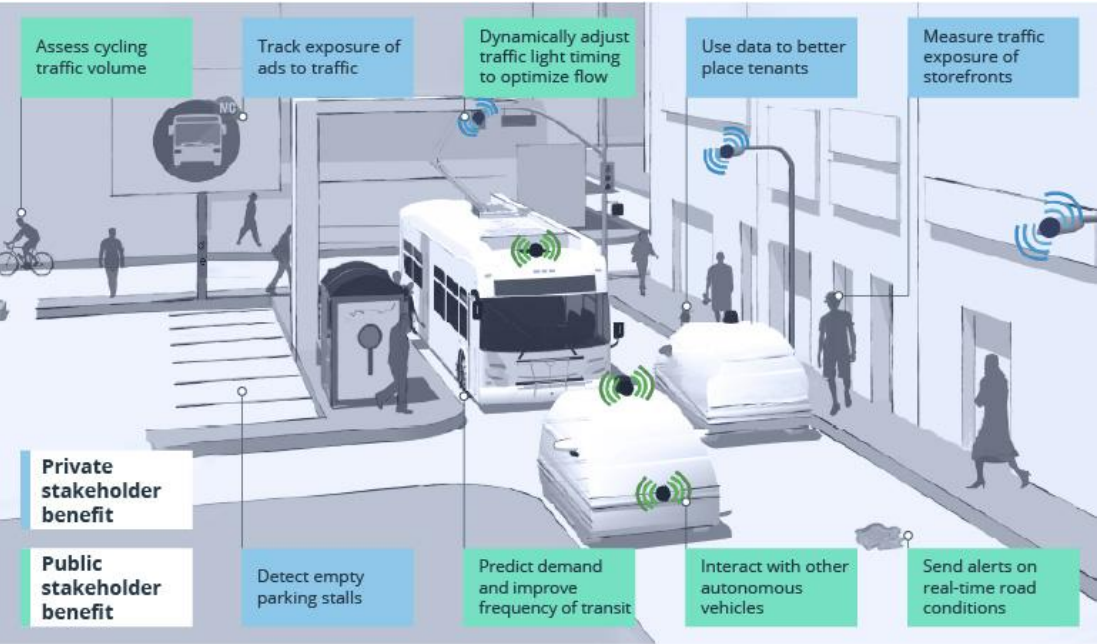
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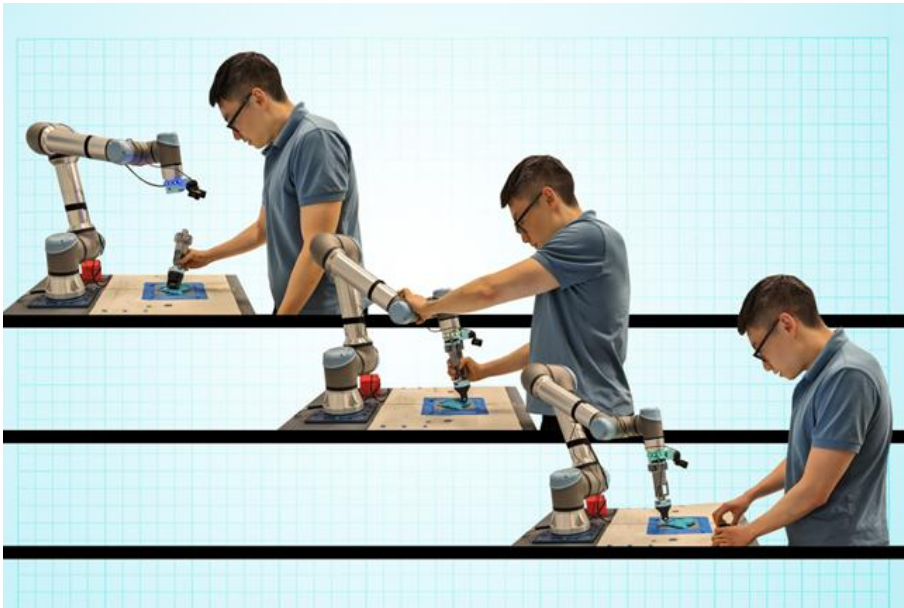
Source: www.datasciencedojo.com



Source: www.youtube.com

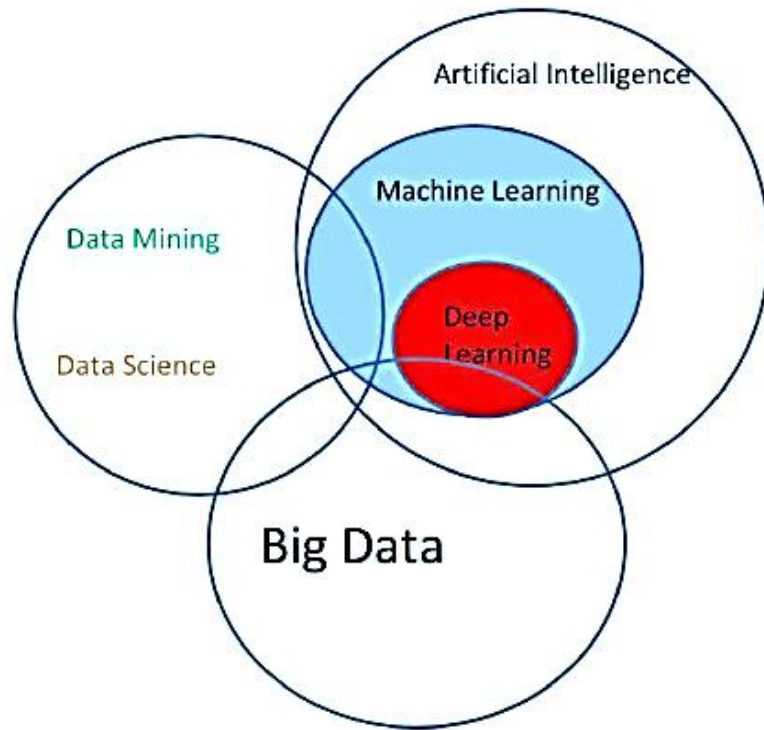


Source: www.medium.com



Source: www.freepik.com

From Data to AI



Source: www.argility.com

Foundation: It all starts with **Data** (Big Data, Data Mining) as the essential fuel.

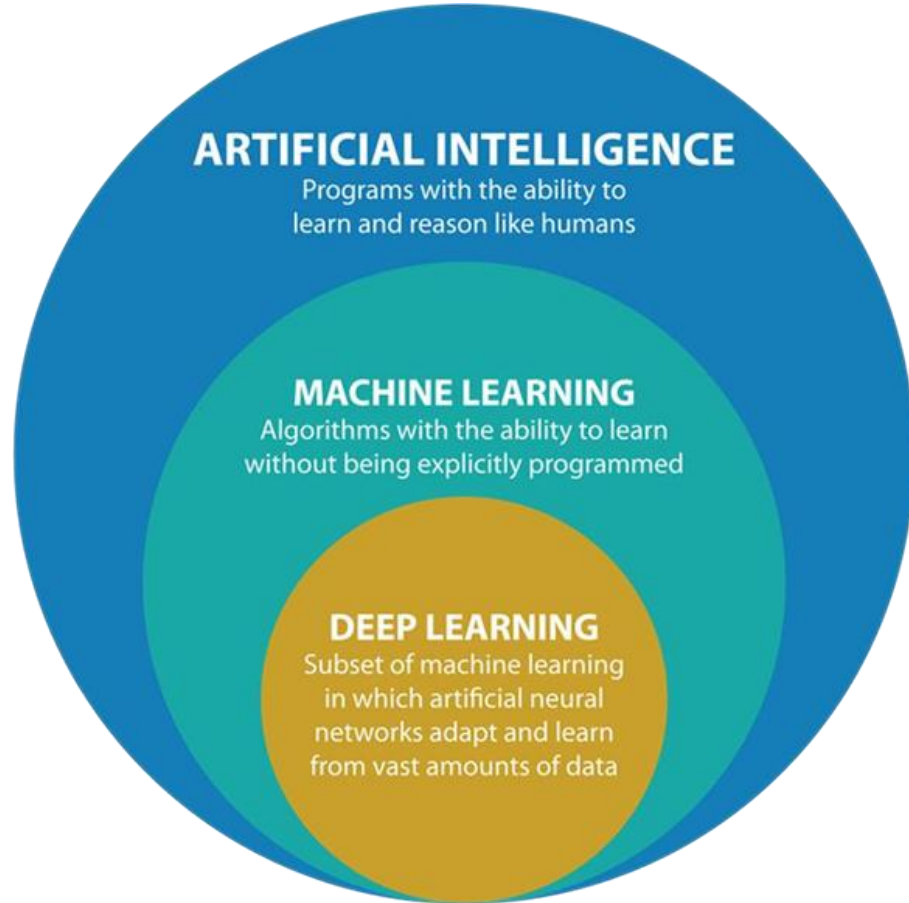
The Goal: The overarching field is **Artificial Intelligence (AI)**, creating intelligent machines.

Key Method: A primary method for achieving AI is **Machine Learning**, where algorithms learn from data.

Advanced Technique: A powerful subset of Machine Learning is **Deep Learning**, which uses complex neural networks.

The Umbrella: The entire process is often involving **Data Science**, which uses scientific methods to extract insights from data.

AI to Deep Learning



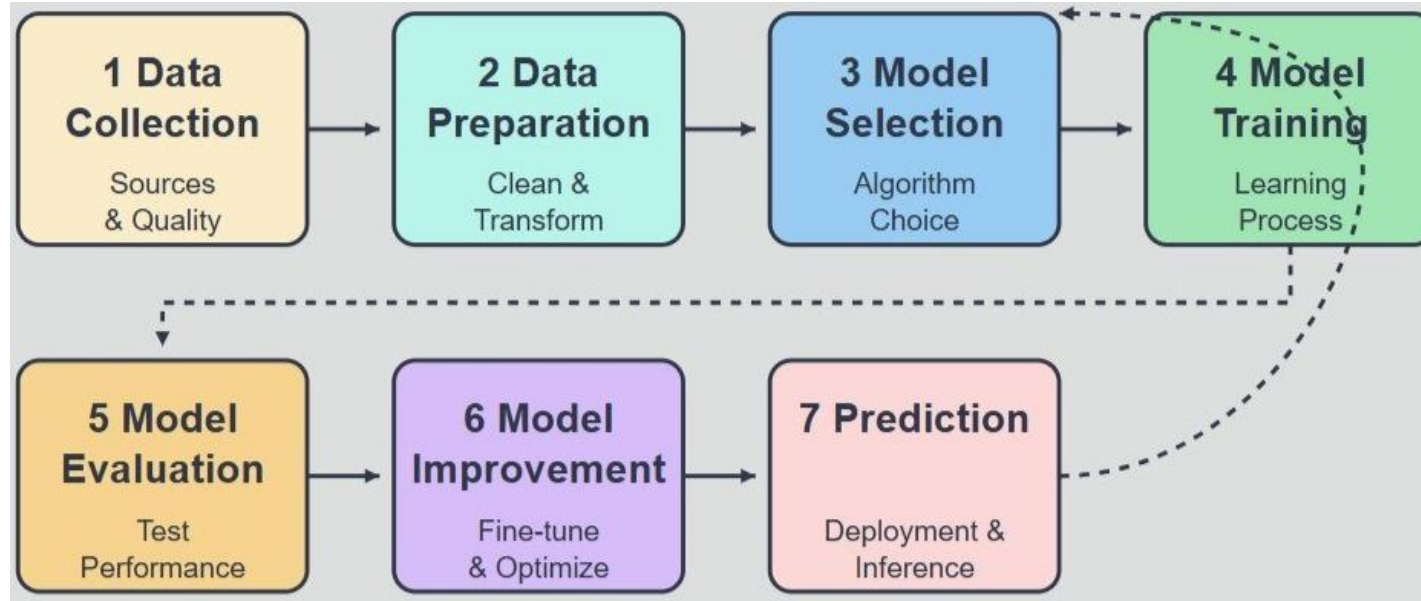
Source: www.argility.com

Hierarchical Relationship: Deep Learning is a specialized branch of Machine Learning, which itself is a key method for achieving the broader goal of Artificial Intelligence.

Increasing Specialization: As you move from AI to ML to Deep Learning, the focus narrows from a general goal of human-like intelligence to specific, data-driven techniques using complex neural networks.

What is Machine Learning?

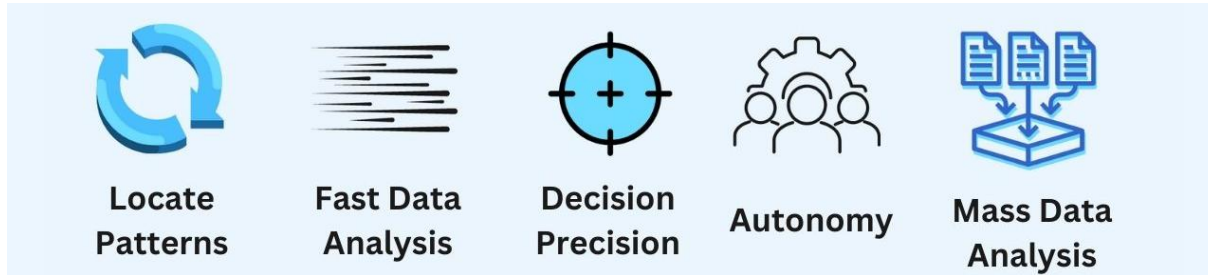
Machine Learning (ML) is defined as the subfield of Artificial Intelligence that focuses on the development of algorithms that have the ability to learn from data without being explicitly programmed for every single rule.



Source: [Kharche, 2025](#)

- A subset of AI focused on systems that learn from data.
- Instead of being programmed explicitly, ML models find patterns and make predictions.
- The goal: Learn from data to improve automatically.
- Combines statistics, mathematics, and optimization with computing power.

Machine Learning Applications



Source: www.argility.com

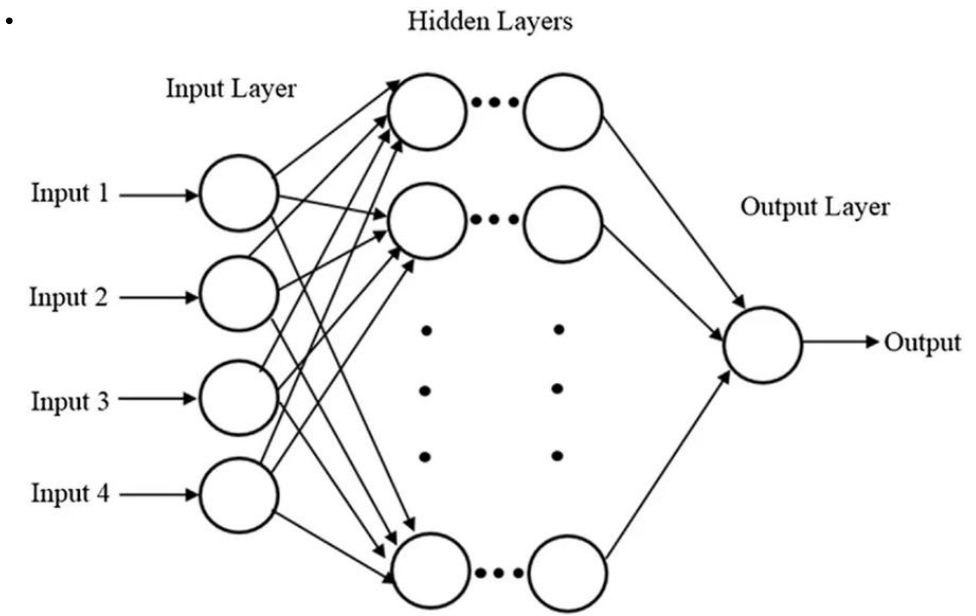
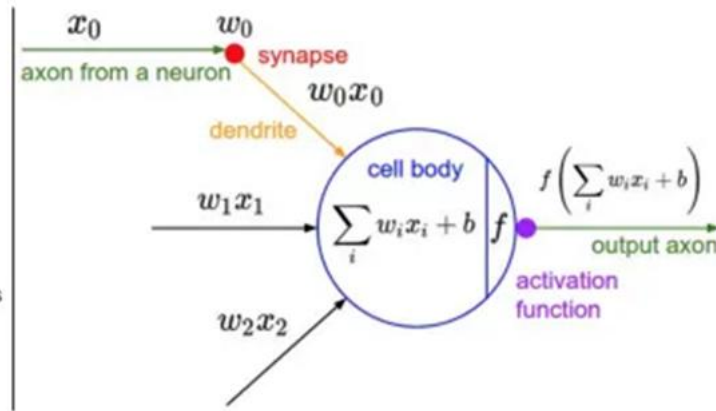
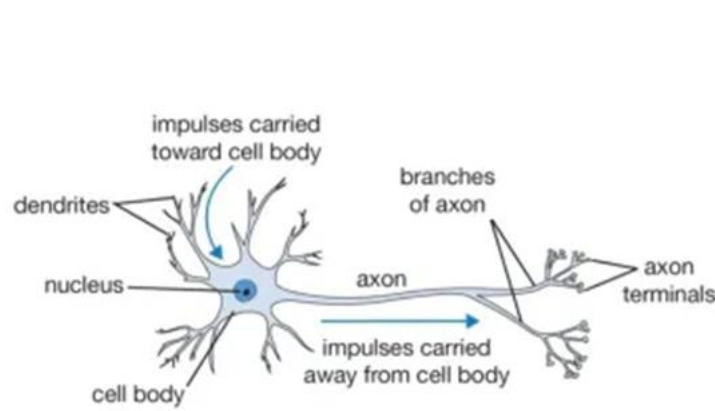
- Identifies hidden patterns and trends
- Processes information at high speed
- Enables highly accurate, data-driven decisions
- Allows systems to operate with self-governance
- Analyses massive volumes of data efficiently



Source: www.argility.com

What is Deep Learning?

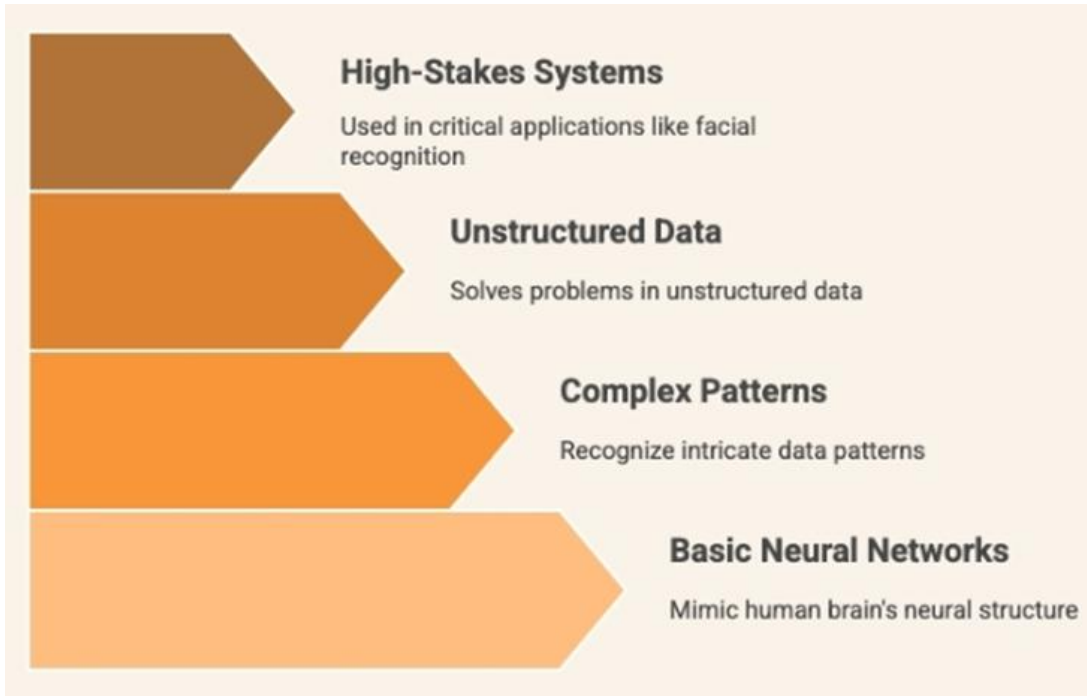
Deep Learning (DL) is a subset of machine learning that uses multi-layered artificial neural networks to automatically learn and make decisions from vast amount of data.



Source: www.mplsvpn.info/

- Mimics the human brain's structure using interconnected neural networks
- Excels at processing complex, unstructured data like images, sound, and text
- Automatically discovers the features needed for classification, reducing the need for manual input
- It powers advanced technologies like facial recognition, self-driving cars, and intelligent assistants

Characteristics of Deep Learning

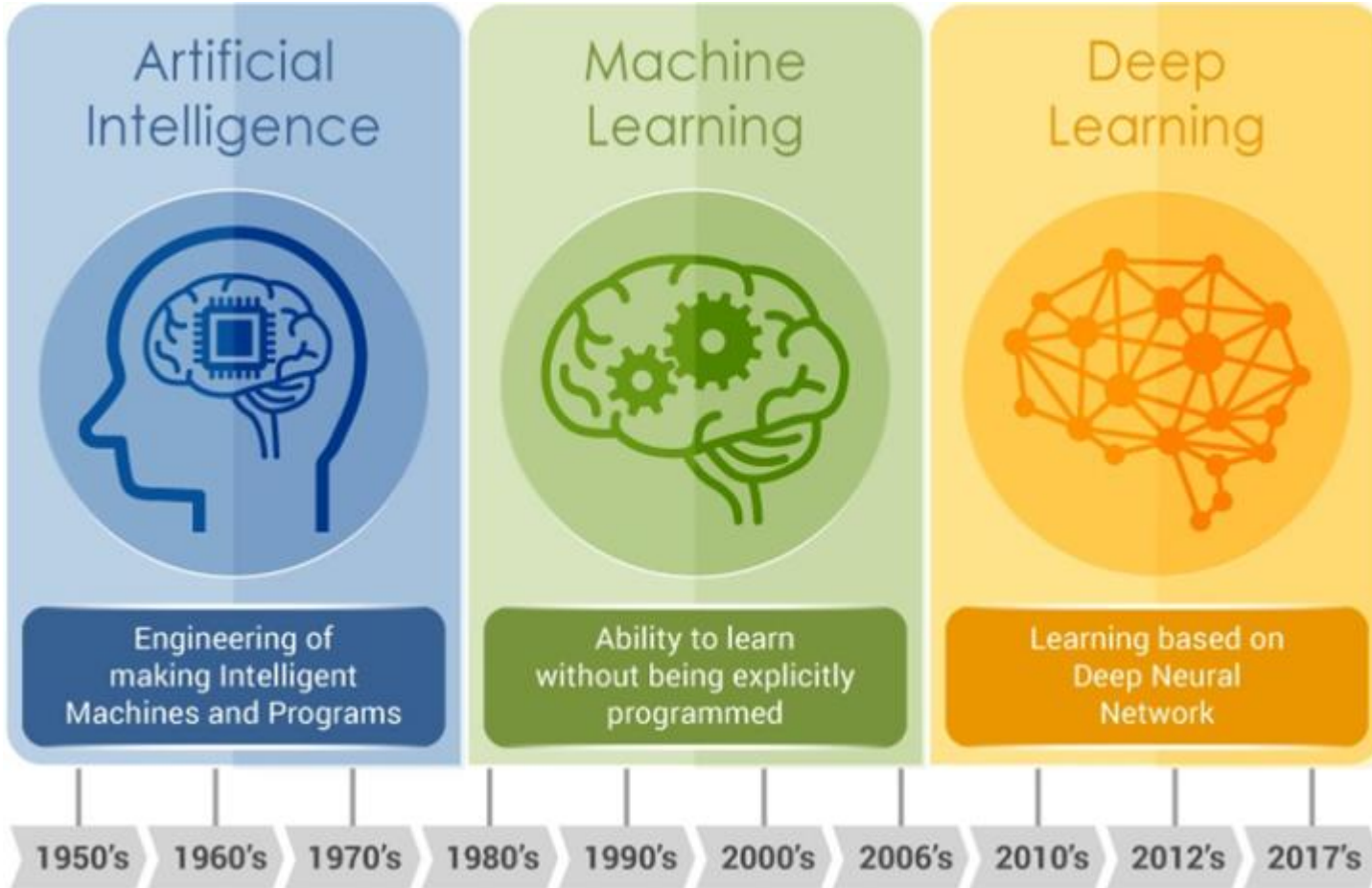


Source: www.morsoftware.com



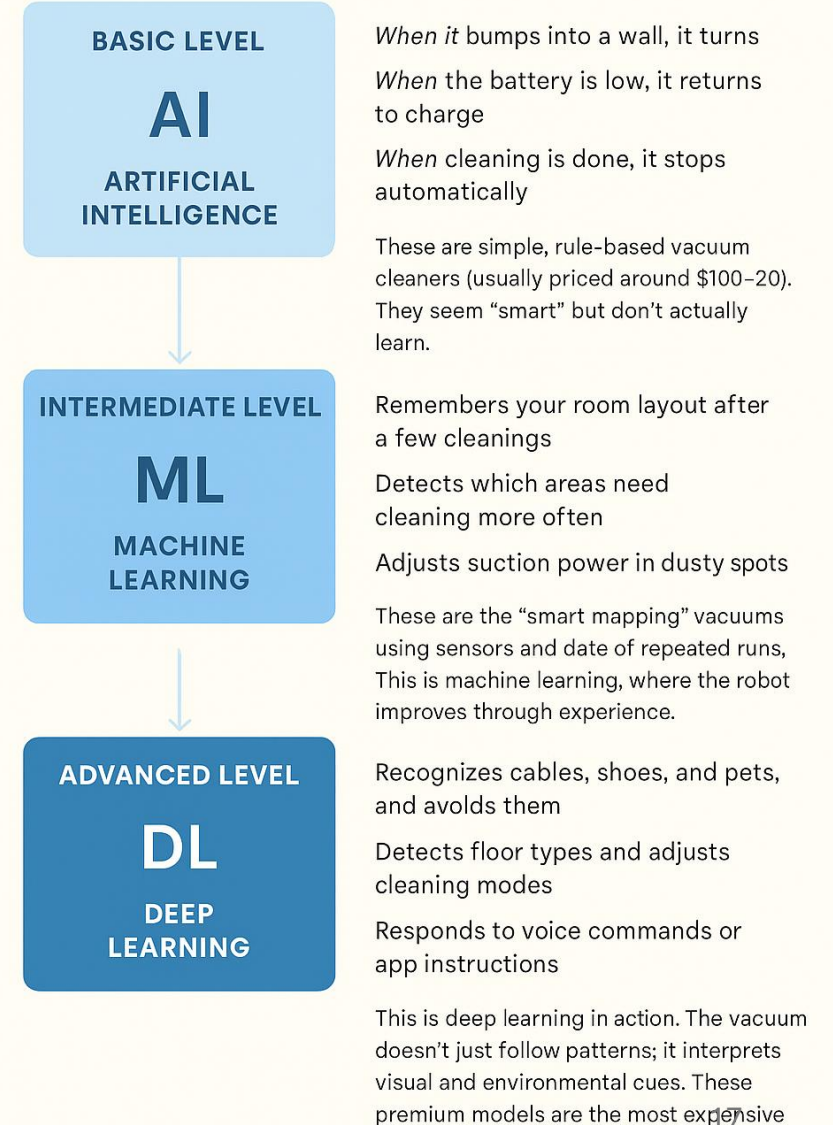
Source: www.morsoftware.com

AI, ML, DL: Know the Difference



Source: www.medium.com

LEVELS OF INTELLIGENCE IN SMART VACUUM CLEANERS



AI, ML, DL: Know the Difference

Feature	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Scope & Definition	Broad field focused on creating systems that mimic human intelligence.	Subfield of AI that enables systems to learn automatically from data.	Specialized area of ML that uses multi-layered neural networks.
Goal	Simulate human-like reasoning and decision-making.	Learn from data to improve performance on specific tasks.	Achieve end-to-end autonomous learning from raw data.
Approach	Based on rules, logic, and symbolic reasoning.	Uses algorithms that identify patterns in data.	Uses deep neural networks for hierarchical feature learning.
Data Requirements	Works with small or symbolic datasets.	Requires structured and labeled data in moderate amounts.	Needs very large volumes of labeled or unstructured data.
Hardware Needs	Runs efficiently on standard CPUs.	May benefit from GPUs for faster processing.	Relies heavily on GPUs for training and inference.
Feature Engineering	Manual and knowledge-driven.	Requires domain expertise for feature selection.	Features are learned automatically by the network.
Training Time	Short or moderate depending on complexity.	Moderate, increases with data size.	Long due to complex models and massive datasets.
Use Cases	Rule-based systems, chatbots, gaming, robotics.	Fraud detection, recommendation systems, predictive analytics.	Image recognition, speech processing, autonomous driving, language models.

THANK YOU

Any Questions?

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